

Question ID: e336a1d2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A cube has an edge length of **41** inches. What is the volume, in cubic inches, of the cube?

Answer

- A. **164**
- B. **1,681**
- C. **10,086**
- D. **68,921**

Correct Answer: D

Rationale

Choice D is correct. The volume, V , of a cube can be found using the formula $V = s^3$, where s is the edge length of the cube. It's given that a cube has an edge length of **41** inches. Substituting **41** inches for s in this equation yields $V = 41^3$ cubic inches, or $V = \mathbf{68,921}$ cubic inches. Therefore, the volume of the cube is **68,921** cubic inches.

Choice A is incorrect. This is the perimeter, in inches, of the cube.

Choice B is incorrect. This is the area, in square inches, of a face of the cube.

Choice C is incorrect. This is the surface area, in square inches, of the cube.